

MIAE 215 · Mini-Course Lesson 6

Fixed-Size Arrays & Numerical Data Buffers

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Source: [Lesson 6 Web Page](#) | Video: [lesson6_arrays.mp4](#)

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1. Concept & Continuous Memory Architecture

An **array** is a contiguous block of memory allocated to store multiple elements of the identical data type under a single identifier.

Array 'double sensor_readings[4]' in RAM ($4 * 8 = 32$ contiguous bytes):

Index 0	Index 1	Index 2	Index 3
[12.4 V]	[14.1 V]	[11.8 V]	[13.5 V]

Address: 0x100 Address: 0x108 Address: 0x110 Address: 0x118

2. Declaration & Zero-Based Indexing

```
int counts[5]; // Allocates 5 integers: counts[0] through counts[4]

// Initializing elements:
counts[0] = 10;
counts[1] = 25;
counts[2] = 40;
counts[3] = 55;
counts[4] = 70;
```

3. The Out-of-Bounds Memory Corruption Trap

In standard C++, **there is no automatic array bounds checking!**

```
int arr[5]; // Valid indices: 0, 1, 2, 3, 4
arr[5] = 99; // CATASTROPHIC BUG: Overwrites adjacent memory outside the array!
```

Writing past the end of an array can overwrite other variables, corrupt return pointers, or cause runtime segmentation faults.

4. Statistical Computations on Arrays

Calculating Mean & Sum:

```
double readings[5] = {10.5, 12.2, 11.0, 13.8, 12.5};
double sum = 0.0;

for (int i = 0; i < 5; i++) {
    sum += readings[i];
}

double average = sum / 5.0;
cout << "Average Sensor Reading: " << average << " V
";
```

5. Teacher Code & Exercise Walkthroughs

- [lesson6.cpp](#) : Demonstration of array declaration, indexing, and loops.
- [lesson6_exercises.cpp](#) & [lesson6_exercises_solutions.cpp](#) : Exercise questions and solutions.
- [lesson6_more.cpp](#) & [lesson6_more_solutions.cpp](#) : Supplemental practice exercises.